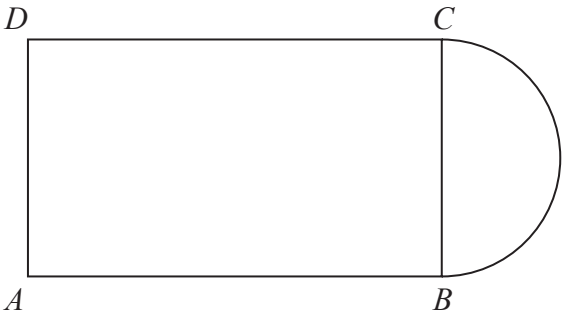
5 The diagram shows a shape made from a rectangle and a semicircle.



The perimeter of the rectangle is 64 cm.

the length of AB : the length of $BC = 3 : 1$

(a) Work out the area of the shape.
Give your answer correct to 3 significant figures.

AB + BC = 64 ÷ 2 = 32 cm

1 mark
32 cm is four ratio parts.

1 ratio part = 32 ÷ 4 = 8 cm

AB = 24 cm; BC = 8 cm

1 mark
Use ratio 3 : 1.

Rectangle area = 24 × 8 = 192 cm^2

1 mark
Multiply the sides.

Radius = 8 ÷ 2 = 4 cm

Semicircle area = ½ × π × 4^2 = 8π cm^2

1 mark
Half of πr^2, with r = 4 cm.

Total = 192 + 8π = 217.1327... cm^2

1 mark
Add the areas; round to 3 significant figures.

217 ..... cm^2

(5)

The ratio is now reversed.

the length of AB : the length of $BC = 1 : 3$

(b) How does this affect the area of the shape?
You must give a reason for your answer.

Area increases. The rectangle area remains 24 × 8.

The semicircle diameter becomes 24 cm, rather than 8 cm.

Its radius triples, so its area becomes nine times as large.

(1)

1 mark
The rectangle is unchanged; the semicircle grows.

(Total for Question 5 is 6 marks)

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